

Community Acquired Pneumonia

Diagnosis

Classification

Management

Duration

CASES

Diagnosis

Classification

Management

Duration

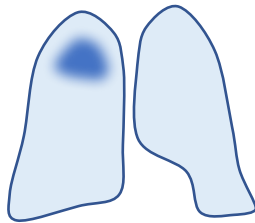
CASES

Diagnosis and Microbiology

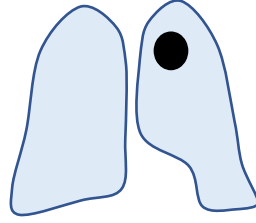
Signs/Symptoms

Fever, cough, dyspnea

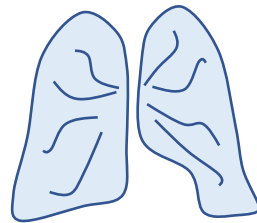
Radiologic (CXR or CT)
findings consistent with
pneumonia



Lobar/ focal
consolidation



Cavitary lesion



Interstitial
pattern

Microbiology

GPCs

S. pneumoniae
S. aureus

GNRs

H. influenzae
M. catarrhalis

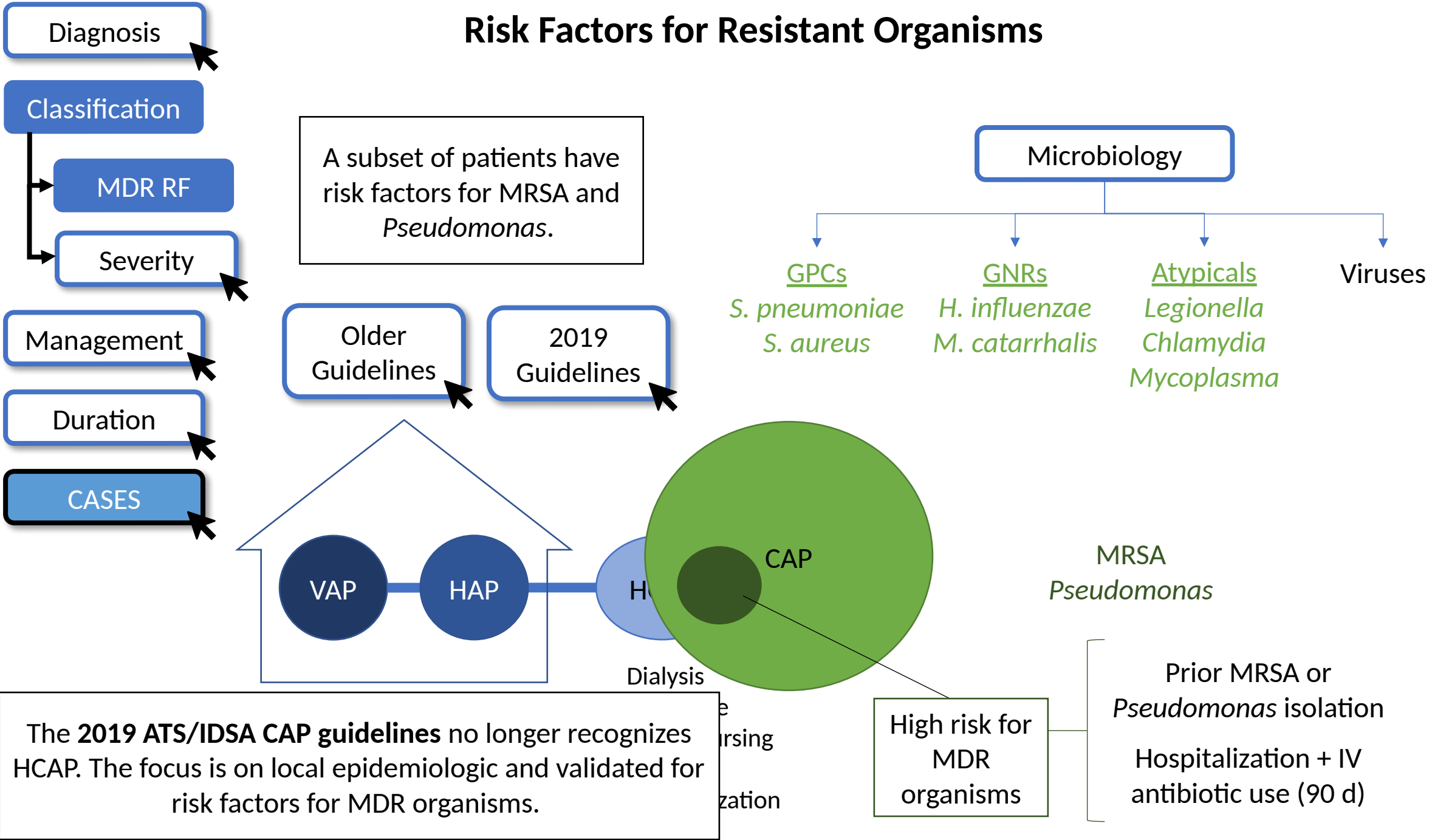
Atypicals

Legionella
Chlamydia
Mycoplasma

Viruses

Influenza
COVID-19

Risk Factors for Resistant Organisms



Pneumonia Severity Scoring Systems

PSI or CURB-65 can be used to determine location of treatment

Diagnosis

Classification

MDR RF

Severity

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PSI

- Demographics
- Comorbidities (cancer, liver/renal/heart)
- Exam (AMS, RR ≥ 30 , SBP < 90 , $\uparrow$ or $\downarrow$ T, HR ≥ 125)
- Lab/Radiology (effusion, SaO₂ $< 90\%$)

CURB-65

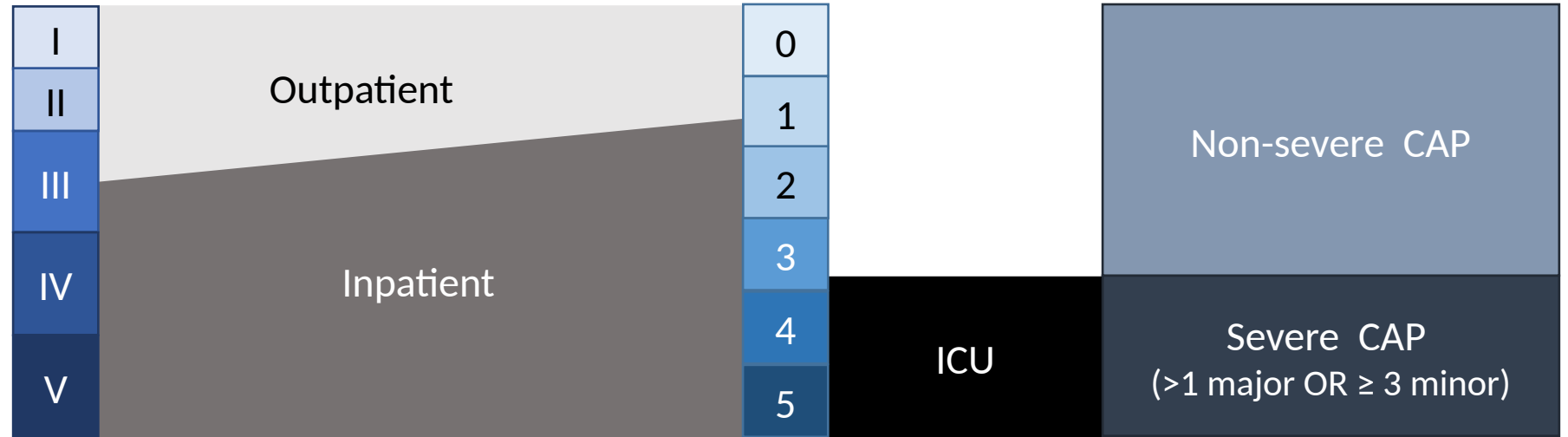
Confusion
Urea > 20 mg/dL
RR ≥ 30
BP < 90
Age ≥ 65

ATS Criteria

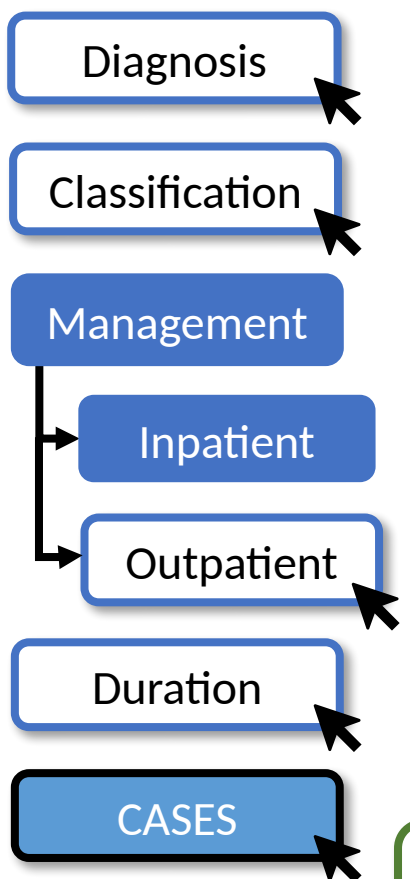
Major
Septic shock
MV

Minor
P:F < 250 RR > 30
BUN > 20 T $< 36^{\circ}\text{C}$
WBC $< 4\text{k}$ Confusion
Plt $< 100\text{k}$ Multi-lobar
 $\downarrow$ BP

Severity

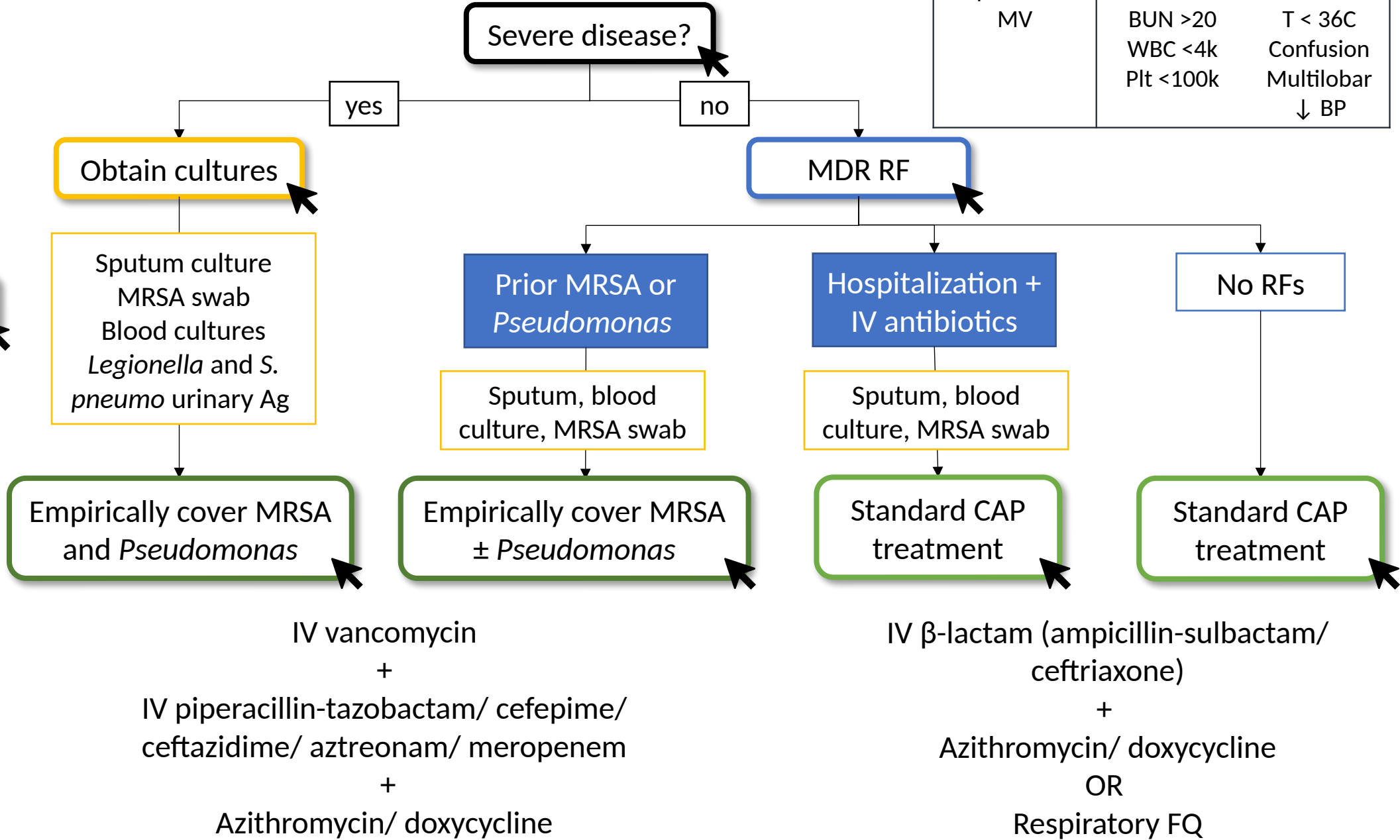


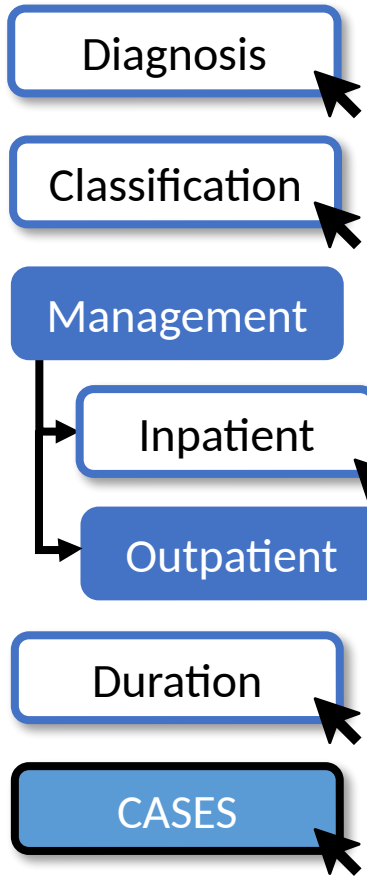
Site of management



Inpatient Evaluation of CAP

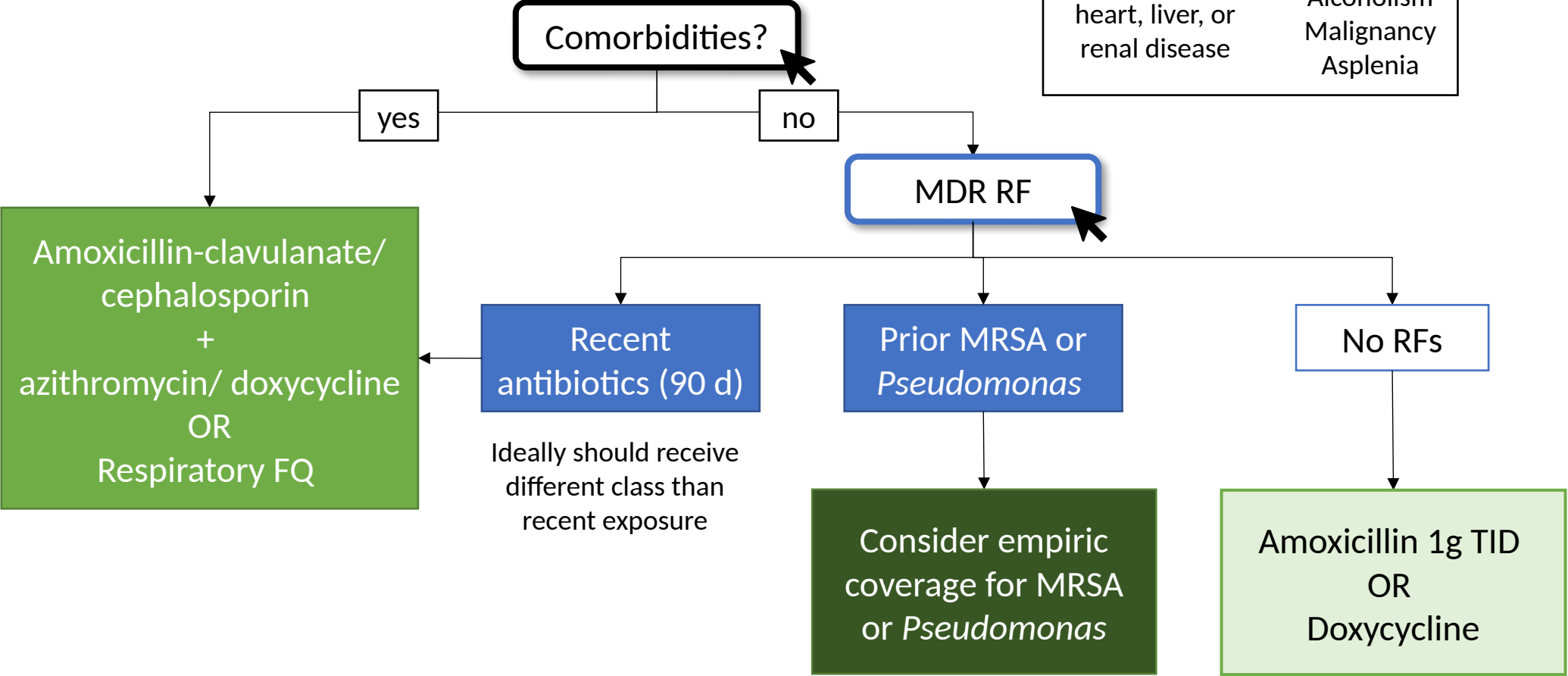
<u>Major (1)</u>	<u>Minor (≥3)</u>	
Septic shock	P:F <250	RR > 30
MV	BUN >20	T < 36C
	WBC <4k	Confusion
	Plt <100k	Multilobar
		↓ BP





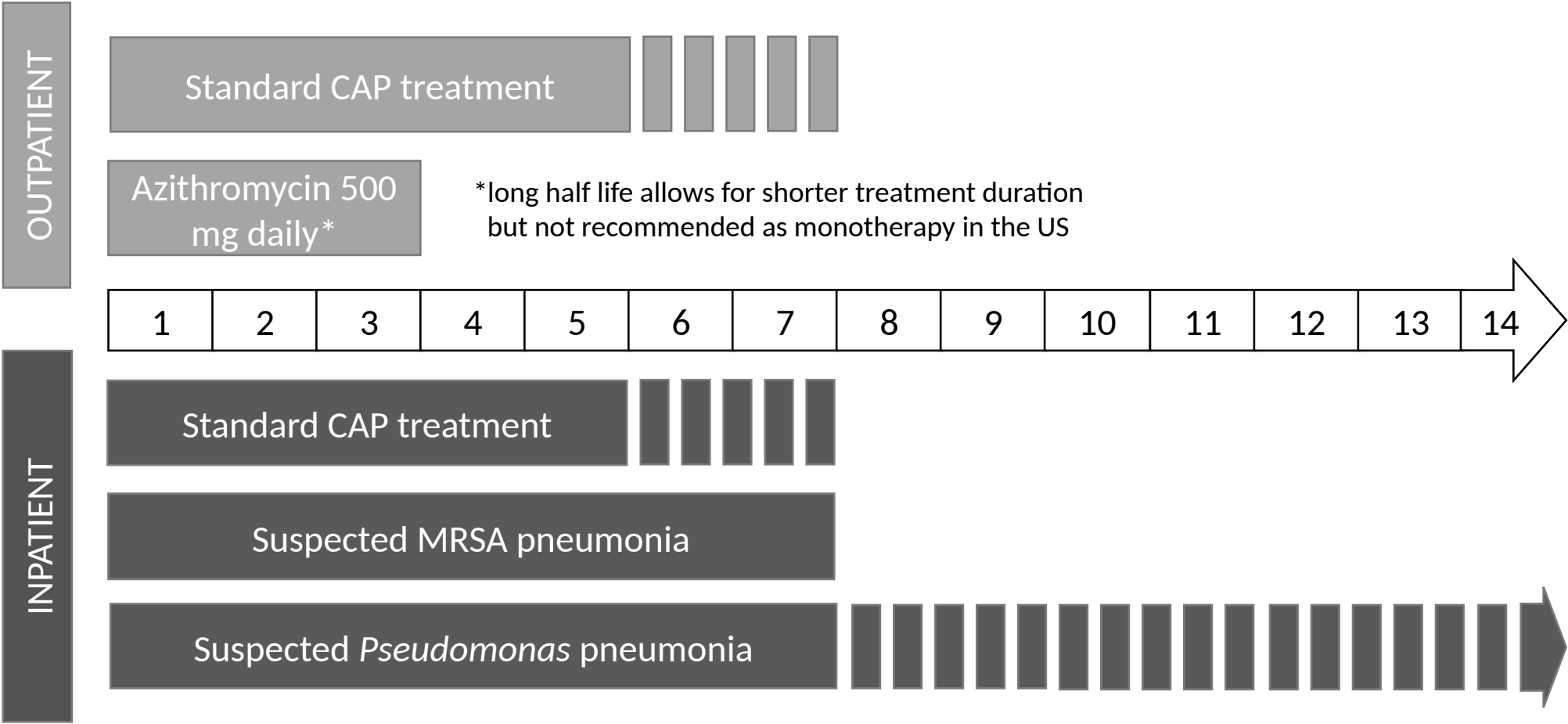
Outpatient Evaluation of CAP

- Chronic lung,
heart, liver, or
renal disease
- DM
Alcoholism
Malignancy
Asplenia



- Diagnosis
- Classification
- Management
- Duration
- CASES

Duration of Treatment



Diagnosis

Classification

Management

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CASES

Case 1

70 yo presents to ED with 4 d of productive cough, malaise and fevers. T 100.9F, HR 110, BP 99/53, RR 33, SpO2 90%.

WBC is 17k, lactate is 2.1. CXR is notable for a focal consolidation in the RLL.

Inpatient. CURB-65 score is 2 (age, and RR >30). PSI is preferred to CURB-65 to determine site of management.

***S. pneumo*, GNRs (*H. flu*, *Moraxella*), atypicals.** Viruses are also a major cause of CAP.

None. In the absence of risk factors for resistant organisms or severe pneumonia, blood and sputum cultures are not needed.

Ceftriaxone + azithromycin.
Doxycycline can be used for atypical coverage when there are QT concerns.

5-7 d. Treatment duration should be guided by clinical stability (e.g., resolution of VS).

Case 2

54 yo with DM2 presents with 3 d of fever and cough. He was recently admitted for MRSA cellulitis. T is 103F, HR 122, BP 80/53 after 2 L IVF, RR 33, SpO2 88% on 6L NC. CXR shows a LUL cavitory lesion. COVID-19 and flu PCR is negative.

Inpatient (ICU). Patients with septic shock (not responsive to IVF)

This is severe CAP. **Blood and sputum cultures.** *Legionella/ S. pneumo* urinary Ag.

Procalcitonin is 0.2 µg/L.

Yes. 2019 IDSA/ATS CAP guidelines recommend abx regardless of procalcitonin level. Sensitivity for detecting bacterial CAP is 38-91%.

IV vanc + piperacillin-tazobactam + azithromycin. Severe CAP (and prior MRSA) warrants broad abx. Lung abscess raises suspicion for MRSA, anaerobic, or mycobacteria

Case 3

41 yo with no PMH presents with fever, cough x 2 d. T is 99F, HR 93, BP 110/53, RR 33, SpO2 94%. CXR shows diffuse interstitial infiltrates. Labs show WBC of 12k, BUN 12, Cr 1.1. COVID-19 and flu PCR is negative.

Outpatient. CURB-65 score is 0, PSI class I.

Amoxicillin 1 g TID for 5 d.
Doxycycline 100 mg BID is an option.

Amoxicillin-clavulanate/ cefpodoxime + azithromycin.
Patients with comorbidities should receive broader antibiotic coverage.

Respiratory FQ (levofloxacin/ moxifloxacin). Recent antibiotic use broader coverage, ideally with a different class.

2023-2025 Practice Updates - CAP

2025 ATS CAP guideline update (Am J Respir Crit Care Med 2025):

Corticosteroids - NEW: give for SEVERE CAP (hydrocortisone <400 mg/day x5-7 d) per CAPE COD (NEJM 2023; mortality RR 0.62, NNT ~17). The original deck said "against steroid use" - that applies to NON-severe CAP; use dexamethasone if COVID-19.

Shorter courses reaffirmed - 3-5 days for patients clinically stable by 48-72 h; never <5 d unless procalcitonin-guided per protocol.

More selective empiric broad-spectrum coverage - cover MRSA/Pseudomonas only with validated risk factors; MRSA nares PCR negative supports stopping vancomycin.

Lung ultrasound endorsed as an alternative imaging tool for diagnosis.

Unchanged (verified current): HCAP remains retired (2019); standard CAP = beta-lactam + macrolide (or respiratory FQ); PSI preferred over CURB-65; procalcitonin NOT used to start antibiotics; 5-7 day duration.

Take-Home Points

CAP pathogens: *S. pneumoniae*, GNRs (*H. flu*, *Moraxella*), atypicals, and viruses. MDR risk = prior MRSA/*Pseudomonas* OR hospitalization + IV antibiotics within 90 days.

Use PSI (preferred) or CURB-65 for site of care; use ATS major/minor criteria for ICU admission (>1 major or ≥3 minor).

Severe CAP and prior MRSA/*Pseudomonas*: culture and empirically cover MRSA +/- *Pseudomonas*. Non-severe CAP without risk factors: do not culture.

Standard inpatient CAP: IV beta-lactam (ampicillin-sulbactam/ceftriaxone) + azithromycin/doxycycline, or a respiratory fluoroquinolone.

Outpatient: stratify by comorbidities and MDR risk. No comorbidities: amoxicillin 1 g TID or doxycycline. Treat 5-7 days, guided by clinical stability.

NEW 2025: add corticosteroids for SEVERE CAP; do not give in non-severe CAP.

Attribution & Changelog

Adapted from TeachIM "Community Acquired Pneumonia" (teachim.org), published Sept 2021.

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Source: teachim.org/teaching_material/cap/ | Deck: CAP-Learner-Presentation.pptx

Corrections (this optimization, 2026-06-10):

Slide 7 (Duration): azithromycin "500 mg TID" -> "500 mg daily" (azithromycin is once-daily; long half-life = shorter, once-daily course). FLAGGED for human review (regimen inferred).

Slide 7: "INTPATIENT" -> "INPATIENT" (spelling).

Slide 8 (Case): outpatient "Amoxicillin 500 mg TID" -> "1 g TID" to match the deck slide 6 standard and IDSA 2019 high-dose. FLAGGED for human review (harmonized value).

Added: 2023-2025 practice-update slide, take-home slide, this attribution slide. All original click-to-reveal animations and graphics preserved.

Companion: Teaching-Script-CAP-1.docx - reviewed, no changes. Newer context source-cited; clinician-review before teaching.